

Optimization of Mean and Standard Deviation of Multiple Responses Using Patient Rule Induction Method

Jin-Kyung Yang, Hanyang University, Seoul, South Korea

Dong-Hee Lee, Hanyang University, Seoul, South Korea

ABSTRACT

In product and process optimization, it is common to have multiple responses to be optimized. This is called multi-response optimization (MRO). When optimizing multiple responses, it is important to consider variability as well as mean of the multiple responses. The authors call this problem as extended MRO (EMRO) where both of mean and variability of the multiple responses are optimized. In this article, they propose a data mining approach to EMRO. In these days, analyzing a large volume of operational data is getting attention due to the development of data processing techniques. Traditional MRO methods takes a model-based approach. However, this approach has limitations when dealing with a large volume of operational data. The authors propose a particular data mining method by modifying patient rule induction method for EMRO. The proposed method obtains an optimal setting of the input variables directly from the operational data where mean and standard deviation of multiple responses are optimized. The authors explain a detailed procedure of the proposed method with case examples.

KEYWORDS

Data Mining, Design of Experiments, Desirability Function, Multi-Response Optimization, Operational Data, Patient Rule Induction Method, Process Optimization, Response Surface Methodology

1. INTRODUCTION

Response surface methodology (RSM) consists of a group of techniques used in empirical study between a response and a number of input variables. The researcher attempts to find the optimal setting for the input variables that either maximizes or minimizes the response (Myers et al., 2009). RSM assumes that variance of the response is constant and focuses on optimizing the mean of the response. However, the constant variance assumption might not be valid in practice. In such cases, not only the mean response, but also the standard deviation of the response should be considered in determining the optimum conditions for the input variables.

Dual-response optimization (DRO) attempts to optimize both the mean and standard deviation of the response. The conventional approach to DRO requires building statistical models for mean and

DOI: 10.4018/IJDWM.2018010104

Copyright © 2018, IGI Global. Copying or distributing in print or electronic forms without written permission of IGI Global is prohibited.

standard deviation of the response, by fitting response surface models (usually second-order models) to experimental data. Then, an optimal setting for the input variables is obtained by analyzing the statistical models (Lee et al., 2010; Lee & Kim, 2012).

Usually, RSM focuses on optimization of a single response. However, it is very common to consider several response variables in product and process development. This is called a multi-response problem. Some typical problems include the tire tread compound problem (Derringer & Suich, 1980), the injection molding of a washing machine agitator problem (Reddy et al., 1997), the rapid prototyping system problem (Palmer et al., 2006), and semiconductor manufacturing process problems (Lee & Kim, 2012; Lee & Kim 2013). Similar to DRO, multi-response optimization (MRO) usually builds statistical models to analyze the means of multiple responses by fitting response surface models (usually second-order models) to experimental data. Then, an optimal setting for the input variables is obtained by analyzing the statistical models (Lee & Kim, 2012).

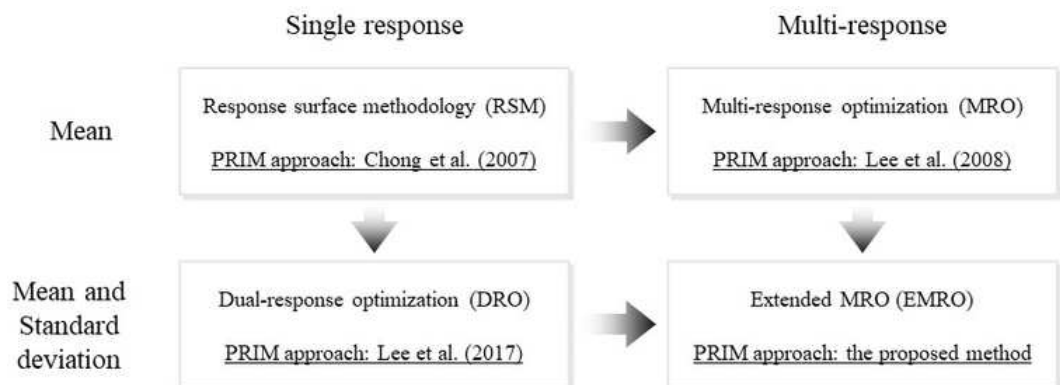
There have been many methods for MRO. However, most of them consider only the mean of multiple responses, like the traditional RSM. As previously mentioned, the constant variance assumption might not be valid in practice. Thus, methods for optimizing both mean and standard deviation of multiple responses are needed. We call this research area extended MRO (EMRO). Figure 1 shows the relationships among RSM, DRO, MRO, and EMRO. In spite of the importance of EMRO, limited studies have been performed.

One possible approach to EMRO is to utilize the conventional modeling methods of RSM, DRO, and MRO. However, these approaches have two practical limitations. First, efforts for conducting experiments can be a burden for process engineers. In order to build the statistical models, experiments are conducted and experimental data are gathered. In particular, in the case of DRO, a large number of experiments are usually conducted, because replicate runs at each design point are needed to fit the statistical model for standard deviation of the response, whereas replicate runs are not mandatory in RSM.

Second, the optimal settings for the input variables obtained from the conventional approaches have uncertainty (Lee & Lee, 2016). This is because the optimal setting is obtained from the prediction models. Care must be taken regarding uncertainty when the prediction capability of the models is poor. As the prediction model varies from the true model, the resulting optimal setting might be quite far from optimal (Xu & Albin, 2003). Also, it is reported that the prediction capability of the models is often poor when the models are fitted to operation data from manufacturing lines, whose volume is typically large (Lee & Kim, 2008).

At present, it is common to have operational data rather than experimental data, and as a result, analyzing the operational data becomes more important. This phenomenon has become very common

Figure 1. Relationships among RSM, DRO, MRO, and EMRO



due to the development of information technology, such as IoT (internet of things) in manufacturing lines. In this environment, data from equipment in manufacturing lines are gathered automatically by sensors and then stored.

When a large volume of operational data, such as data from manufacturing lines, are available, a data mining approach is a good alternative for resolving the difficulties of the conventional approaches. The data mining approach does not use a prediction model and so is free of uncertainty issues. Also, it utilizes operational data; thus, no experiments are required. One attractive data mining approach to process optimization is the patient rule induction method (PRIM) (Friedman & Fisher, 1999). PRIM searches a set of sub-regions of the input variable space within which the performance of the response is considerably better than that of the entire input domain (Chong et al., 2007).

PRIM has been adopted for process optimization in some studies. Chong et al. (2007) adopted PRIM for process optimization. This work focuses on the mean optimization of a single response variable assuming that the response variable variance is stable. However, the constant variance assumption might not valid in practice. Lee & Kim (2008) suggested a multi-response PRIM, called MR-PRIM, which extended Chong et al. (2007)'s method by adopting PRIM for MRO. MR-PRIM also neglects the optimization of the response variables. Recently, Lee et al. (2017) suggested dual response PRIM, called DR-PRIM, which adopted PRIM for optimizing mean and standard deviation of a single response. Although DR-PRIM considers variability, it has limited applicability because it only considers a single response variable. These PRIM-based approaches are shown in Figure 1.

Overall, no works have adopted PRIM for EMRO. We believe that PRIM is an attractive alternative for EMRO because it can resolve practical difficulties of the conventional modeling approaches. With operational data, it is often not possible to build a statistical model of standard deviation of the response variable because standard deviation can only be estimated when replicate data is available (Lee et al., 2017). In PRIM, replicates are not required because it does not build statistical models.

The purpose of the proposed method is to optimize the mean and standard deviation of multiple responses simultaneously by adopting PRIM. The proposed method is a "model-free approach;" thus, it is free from the uncertainty issues resulting from model building. Also, it utilizes operation data and so no effort for creating experimental data is required.

The rest of the paper is organized as follows. In Section 2, the existing DRO and MRO methods and PRIM are reviewed. In Section 3, the proposed method is presented. In Sections 4 and 5, the proposed method is illustrated by a case example with a single response and a case with multiple responses, respectively. Finally, concluding remarks are presented in Section 6.

2. LITERATURE REVIEW

2.1. Dual-Response Optimization

DRO attempts to find a set of input variables that simultaneously optimize the mean and standard deviation of the response. DRO can be formally expressed as follows:

$$\begin{aligned} &\text{Optimize} [\hat{\omega}_{\mu}(\mathbf{x}), \hat{\omega}_{\sigma}(\mathbf{x})] \\ &\text{subject to } \mathbf{x} \in \Omega \end{aligned}$$

where $\mathbf{x}$ is a vector of input variables, and Ω is an experimental region of $\mathbf{x}$. $\hat{\omega}_{\mu}(\mathbf{x})$ and $\hat{\omega}_{\sigma}(\mathbf{x})$ are the estimated mean and standard deviation of the response, respectively. They are often modeled as quadratic functions as follows:

$$\hat{\omega}_{\mu}(\mathbf{x}) = \hat{\beta}_0 + \sum_{i=1}^p \hat{\beta}_i x_i + \sum_{i=1}^p \hat{\beta}_{ii} x_i^2 + \sum_{i < j}^p \sum_{j=1}^p \hat{\beta}_{ij} x_i x_j,$$

$$\hat{\omega}_{\sigma}(\mathbf{x}) = \hat{\gamma}_0 + \sum_{i=1}^p \hat{\gamma}_i x_i + \sum_{i=1}^p \hat{\gamma}_{ii} x_i^2 + \sum_{i < j}^p \sum_{j=1}^p \hat{\gamma}_{ij} x_i x_j$$

DRO was first presented by Myers & Carter (1973) and popularized by Vining & Myers (1990). Vining & Myers (1990) minimized the standard deviation response while keeping the mean response at the specified target. Lin & Tu (1995) suggested a way of minimizing the mean squared error (MSE), which consists of a squared bias component and a variance component:

$$MSE = (\hat{\omega}_{\mu} - T)^2 + \hat{\omega}_{\sigma}^2$$

where T is the target of the mean response.

2.2. Multi-Response Optimization

MRO attempts to find a set of input variables that simultaneously optimize the mean of multiple responses. MRO can be formally expressed as follows:

$$\text{Optimize } [\hat{y}_1(\mathbf{x}), \hat{y}_2(\mathbf{x}), \dots, \hat{y}_m(\mathbf{x})]$$

subject to $\mathbf{x} \in \Omega$

where $\hat{y}_k$ ($k = 1, 2, \dots, m$) is the fitted response function of the k^{th} response variable y_k and m is the number of responses. Like $\hat{\omega}_{\mu}(\mathbf{x})$ and $\hat{\omega}_{\sigma}(\mathbf{x})$, $\hat{y}_k$ is often modeled as a quadratic function.

As mentioned in Section 1, most MRO methods consider only the mean of multiple responses. In order to achieve a more successful process optimization, EMRO needs to be considered where the mean and standard deviation of the multiple responses are optimized.

Most of the existing DRO and MRO methods can be considered as “model-based approaches.” The mean and standard deviation of the response are modeled in the form of a “response surface” based on an RSM framework. However, this model-based approach involves the uncertainty issue described in Section 1. Furthermore, it is not appropriate for large volumes of operational data due to poor prediction capability. Due to these difficulties, we propose a data mining approach to EMRO. In other words, we propose a particular model-free approach for EMRO, called extended multi-response PRIM (EMR-PRIM). The next section briefly reviews PRIM.

2.3. Patient Rule Induction Method

PRIM searches a set of sub-regions of the input variable space within which the performance of the response is considerably better than that of the entire input domain (Chong et al., 2007). Here, the response is considered a larger, better response. Although PRIM originally allows the input variables to have both real and categorical values (Friedman & Fisher, 1999; Chong et al., 2007), only real valued input variables are considered here for the sake of simplicity.

For the p input variables $x_1, x_2, \dots, x_p$, a p -dimensional box B is defined as their intersection:

$$B = s_1 \times s_2 \times \dots \times s_p$$

where s_j is a sub-range of the j^{th} input variable, x_j denoted by $s_j = (l_j, u_j)$ where l_j and u_j are the lower and upper limits, respectively.

Suppose that we have N observations denoted by $\{(y_i, \mathbf{x}_i), i = 1, 2, \dots, N\}$ where y_i and $\mathbf{x}_i$ are values of the response variable and the input variables of the i^{th} observation, respectively. $\mathbf{x}_i$ is a p -dimensional vector denoted by $\mathbf{x}_i = (x_{i1}, x_{i2}, \dots, x_{ip})$. When $\mathbf{x}_i$ is located in box B (i.e., $l_1 \leq x_{i1} \leq u_1, l_2 \leq x_{i2} \leq u_2, \dots, l_p \leq x_{ip} \leq u_p$), it is denoted by $\mathbf{x}_i \in B$.

Given observations $\{(y_i, \mathbf{x}_i), i = 1, 2, \dots, N\}$ there are two statistics that describe the properties of box B . The first is the *support* β_B which denotes the proportion of observations located in box B :

$$\beta_B = \frac{n_B}{N}$$

where n_B denotes the number of observations located inside box B . The support ranges from zero to one. When all of observations are located in box B , $\beta_B = 1$. The second is the box objective, Obj_B , which describes the mean value of the response in box B :

$$Obj_B = \bar{y}_B = \frac{1}{n_B} \sum_{\mathbf{x}_i \in B} y_i$$

The main idea of PRIM is presented below. Given observations $\{(y_i, \mathbf{x}_i), i = 1, 2, \dots, N\}$ an initial box B_0 is formed by defining $l_j = \min_{i=1,2,\dots,N} x_{ij}$, $u_j = \max_{i=1,2,\dots,N} x_{ij}$, $j = 1, 2, \dots, p$. From B_0 $2p$ candidate boxes denoted by $\{C_{01-}, C_{01+}, C_{02-}, C_{02+}, \dots, C_{0p-}, C_{0p+}\}$ are created. The candidate boxes C_{0j-} and C_{0j+} , $j = 1, 2, \dots, p$ are obtained by peeling $100\alpha\%$ of the observations in box B_0 . The parameter α determines peeling rate, and its value is set typically between 0.05 and 0.1. Then, PRIM chooses the option with the largest box objective among the candidate boxes and lets this box be B_1 . If the *support* of B_1 is greater than a stopping parameter, β_0 then PRIM continues to perform more peelings to generate the next box B_2 . Otherwise, the algorithm ends. This iterative process continuous until the *support* of the box becomes less than the predetermined value β_0 . The value of β_0 determines box stability. When β_0 is too small, the box often becomes over-fitted. According to Friedman & Fisher (1999), it is recommended to set between 0.05 and 0.1.

3. PROPOSED EMR-PRIM

In this section, the proposed method, called extended multi-response PRIM (EMR-PRIM) is presented. EMR-PRIM is a modified PRIM, which attempts to optimize the mean and standard deviation of multiple responses. As aforementioned in Section 1, the existing “model-based approaches” have a limitation in that the prediction models cannot effectively explain large operational datasets. The prediction capability of the models is often too low; thus, the obtained setting of the input variables might be far from optimal. In contrast, the proposed EMR-PRIM is a “model-free approach” that avoids the low prediction capability issue.

The basic assumption of the proposed method is that observations for input variables and responses are known. Similar to PRIM in Section 2.3, we have N observations $\left\{(\mathbf{y}_i, \mathbf{x}_i), i = 1, 2, \dots, N\right\}$ where $\mathbf{y}_i$ and $\mathbf{x}_i$ are values of the responses and the input variables of the i^{th} observation, respectively. $\mathbf{y}_i$ is an m -dimensional vector denoted by $\mathbf{y}_i = (y_{i1}, y_{i2}, \dots, y_{im})$, and $\mathbf{x}_i$ is a p -dimensional vector denoted by $\mathbf{x}_i = (x_{i1}, x_{i2}, \dots, x_{ip})$.

Figure 2 shows the four-step pseudo-code of EMR-PRIM. Steps 2, 3, and 4 iterate until the stopping criterion is satisfied. The details of each step are described below.

Step 1. Set Parameter Values: In this step, the values for the two parameters α and β_0 are determined.

Typically, the α value is recommended to be set between 0.05 and 0.1, which is faithful to the original concept “patient” rule induction. Typically, β_0 is set to 0.05. See Friedman & Fisher (1999) for discussion of the β_0 value. Next, the initial box B_0 is defined:

$$B_0 = s_{01} \times s_{02} \times \dots \times s_{0p}$$

$s_{0j} = [l_{0j}, u_{0j}]$ is a subrange of $x_j, j = 1, 2, \dots, p$. Given observations $\left\{(\mathbf{y}_i, \mathbf{x}_i), i = 1, 2, \dots, N\right\}$ subranges are defined by $l_{0j} = \min_{i=1,2,\dots,N} x_{ij}$ $u_{0j} = \max_{i=1,2,\dots,N} x_{ij}, j = 1, 2, \dots, p$. The box objective of box B_0 is defined by:

$$Obj_{B_0} = \sum_{k=1}^m \left(\bar{Y}_{k,B_0} - T_k \right)^2 + s_{k,B_0}^2 \quad (1)$$

where $\bar{Y}_{k,B_0}$ is the mean value of the k^{th} response of the observations located in box B_0 . Because box B_0 contains all N observations in Step 1, the number of observations located in box B_0 n_{B_0} is equal to N , and $\bar{Y}_{k,B_0}$ is the mean of all observations of the k^{th} response:

$$\bar{Y}_{k,B_0} = \frac{\sum_{i=1}^{n_{B_0}} y_{ik}}{n_{B_0}}$$

T_k denotes the target value of the k^{th} response. s_{k,B_0}^2 is the sample variance of the k^{th} response of the observations located in box B_0 s_{k,B_0}^2 is calculated by:

Figure 2. Algorithm of EMR-PRIM

1: Set values of parameters.	Step 1
2: repeat	
3: Generate candidate boxes to be peeled.	Step 2
4: Select the candidate box having the smallest objective value as the box.	Step 3
5: until support < β_0	Step 4

$$s_{k,B_0}^2 = \frac{\sum_{i=1}^{n_{B_0}} (y_{ik} - \bar{Y}_{k,B_0})^2}{n_{B_0} - 1}$$

The box objective consists of mean squared errors (MSE) of all m responses; thus, all responses are involved in the optimization. MSE consists of two terms, one for deviation from the target value and the other for variance. Using MSE as the box objective, the variance is also considered in the optimization. In short, both the mean and variance of the multiple responses are considered during optimization. This is the main difference from the original PRIM.

Step 2. Generate Candidate Boxes to be Peeled: For x_j two candidate boxes denoted by C_{j-} and C_{j+} are generated:

$$C_{j-} = s_{01} \times \cdots \times s_{0j}^{(-)} \times \cdots \times s_{0p},$$

$$C_{j+} = s_{01} \times \cdots \times s_{0j}^{(+)} \times \cdots \times s_{0p}, \quad j = 1, 2, \dots, p$$

where $s_{0j}^{(-)} = [x_{j(\alpha)}, u_{0j}]$ and $s_{0j}^{(+)} = [l_{0j}, x_{j(1-\alpha)}]$. $x_{j(\alpha)}$ is the α percentile of x_j in box B_0 (i.e., $\Pr(x_j < x_{j(\alpha)} | \mathbf{x} \in B_0) = \alpha$ (Lee & Kim, 2008)). A total of $2p$ candidate boxes are generated.

Step 3. Select the candidate box with the smallest box objective as box B_m : The box objectives of the candidate boxes are calculated by Equation (1). A total of $2p$ box objectives are calculated. It should be noted that a small MSE value is preferred in order to locate the candidate box with the smallest box objective value. Therefore, the candidate box with the smallest box objective is selected as box B_1 .

Step 4. Verify That the Stopping Criterion is Satisfied: If the *support* of box B_1 β_{B_1} is smaller than the predetermined threshold, β_0 the algorithm ends with the box B_1 . Otherwise, the algorithm goes to Step 2 and conducts the next peeling from Steps 2, 3, and 4 in order to generate the next box B_2 from B_1 . This iterative procedure continues until the stopping criterion is satisfied.

4. EXAMPLE 1: CONCRETE COMPRESSIVE STRENGTH PROBLEM

In this section, the dual response problem, called the concrete compressive strength problem (Yeh, 1998), is solved by the proposed EMR-PRIM. The response in this problem is concrete compressive strength (y), and it is affected by eight input variables $(x_1, x_2, \dots, x_8)$ such as cement, blast furnace slag, fly ash and so on. The unit of concrete compressive strength is MPa (egapascals). The response is a larger the better type response. More specifically, it is desirable to make a compressive strength be greater than 40 MPa. In this regard, we set the target of the response as 60 MPa ($= T$).

4.1. Optimization of the Concrete Compressive Strength Problem

Step 1. Set the Initial Parameters: Two parameters are set as $\alpha = 0.05$ and $\beta_0 = 0.05$. The initial box B_0 is set as an interaction of the sub-ranges of the eight input variables (i.e., $B_0 = s_{01} \times s_{02} \times \dots \times s_{08}$). The total number of observations, N , is 1030. The sub-range of each input variable is given in Table 1. The mean and variance of observation responses are 35.78 and 280.01, respectively. Thus, the box objective of B_0 is calculated as:

$$Obj_{B_0} = (35.78 - T)^2 + 280.01 = 866.62$$

Step 2. Generate Candidates to be Peeled: For each input variable, a candidate box is generated. A total of 16 candidate boxes $C_{(1-)}, C_{(1+)}, C_{(2-)}, C_{(2+)}, \dots, C_{(8-)}, C_{(8+)}$ are generated. For example, $C_{(3-)}$ and $C_{(3+)}$ are given as below:

$$C_{3-} = s_{01} \times \dots \times s_{03}^{(-)} \times \dots \times s_{08}, C_{3+} = s_{01} \times \dots \times s_{03}^{(+)} \times \dots \times s_{08}$$

where $s_{03}^{(-)} = [x_{3(0.05)}, u_{03}]$ and $s_{03}^{(+)} = [l_{03}, x_{3(0.95)}]$.

Step 3. Select the Candidate Box with the Smallest Box Objective as Box B_m : The box objectives of the 16 candidate boxes are calculated and are:

$$Obj_{C_{1-}} = 832.07, Obj_{C_{2-}} = 862.99, \dots, Obj_{C_{8-}} = 714.45, Obj_{C_{1+}} = 899.77, Obj_{C_{2+}} = 846.41, \dots, Obj_{C_{8+}} = 893.63$$

The box with the smallest box objective is C_{8-} thus, C_{8-} is defined as B_1 . Its box objective Obj_{B_1} is smaller than Obj_{B_0} (i.e., $714.45 < 866.62$). It should be noted that the *support* of B_1 is $\beta_{B_1} = 0.63$. Usually, β_{B_1} is 0.95 when the peeling rate is 0.05. However, in this particular example, 37% of observations were peeled off in the first iteration; thus, β_{B_1} becomes 0.63. This is because many observations have the same value of x_8 thus, they were excluded by a single peeling process.

Step 4. Verify That the Stopping Criterion is Satisfied: Because $\beta_{B_1} > \beta_0$ (i.e., $0.63 > 0.05$), the algorithm returns to Step 2. Then, peeling and candidate box selection are conducted to generate the next box B_2 . This iterative process continues until the *support* becomes less than 0.05. As a result, a total of 21 iterations of Steps 2, 3, and 4 are conducted and shown in the left part of Table 2. Box B_{18} has the smallest objective value and so is selected as the optimal box. The

Table 1. Sub-ranges of input variables

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8
l_{0j}	102	0	0	121.8	0	801.0	594.0	1.0
u_{0j}	540	359.4	200.1	246.9	32.2	1134.3	992.6	365.0

Table 2. Results of training and validation sets obtained by DR-PRIM

Iteration	Training Set			Validation Set		
	Box Obj.	Mean	Variance	Box Obj.	Mean	Variance
1	714.45	39.28	285.20	670.68	39.92	267.47
2	522.85	44.30	276.42	588.62	42.40	278.94
3	419.95	46.93	249.13	498.05	44.82	267.54
4	393.36	47.76	243.54	482.78	45.35	268.12
5	366.63	48.66	238.15	425.59	47.12	259.78
6	327.52	49.77	222.96	385.28	48.45	251.89
7	285.88	50.96	204.24	317.45	50.20	221.36
8	267.28	51.78	199.73	289.60	51.29	213.78
9	250.19	52.42	192.79	265.94	52.79	213.99
10	235.88	53.10	188.32	250.11	53.22	204.19
11	219.93	53.78	181.21	228.68	54.33	196.49
12	205.85	55.51	185.71	186.46	56.58	174.80
13	188.88	56.16	174.13	173.00	57.09	164.52
14	124.37	65.56	93.44	98.41	65.64	66.60
15	110.03	64.88	86.26	110.85	65.63	79.13
16	103.17	64.53	82.68	110.85	65.63	79.13
17	101.12	64.95	76.62	113.40	65.42	84.02
18	97.25	65.14	70.85	138.77	66.48	96.74
19	98.64	65.10	72.65	150.06	66.48	96.74
20	100.52	65.79	67.04	150.06	66.48	96.74
21	98.24	66.77	52.39	150.06	66.48	96.74

mean of B_{18} is 65.14, which is quite close to the target value 60. The box objective of B_{18} is 97.25, which is much smaller than that of B_0 .

4.2. Cross Validation of the Optimization

The obtained optimal box works well for the given data; however, it might not be satisfactory for other data. When the obtained optimal box works well only for the given data, the obtained box can be considered to be over-fitted. In this regard, we conducted cross validation to determine whether the obtained box is over-fitted.

First, we randomly split the existing entire data into a training set and a validation set. Two-thirds of the data became the training set, and the remaining one-third became the validation set. The optimization was conducted with the training set; thus, the optimal box was derived from the training set only. Then, the box objective of the optimal box was recalculated using the validation set data. Table 2 reports the box objective values from the data of training and validation sets. The box objective from the validation set is slightly larger than the training set; however, the difference is not significant. This indicates that the obtained optimal box is not over-fitted.

5. EXAMPLE 2: CONCRETE COMPRESSIVE STRENGTH PROBLEM WITH TWO RESPONSES

In this section, the proposed EMR-PRIM is illustrated via a case example with two responses, concrete compressive strength (y_1) and concrete tensile strength (y_2). We arbitrarily generated y_2 values to illustrate the proposed method. The target values of the two responses are 60 ($= T_1$) and 50 ($= T_2$), respectively. The purpose of this case study is to maintain the means of y_1 and y_2 at their target values, while minimizing the standard deviation of responses by controlling the input variables.

5.1. Optimization of the Concrete Compressive Strength Problem with Two Responses

In Step 1, initial parameters are determined. Two parameters are set as $\alpha = 0.1$ and $\beta_0 = 0.05$. The total number of observations and the sub-ranges of input variables are the same as described in Section 4. The mean and variance of observation responses are 35.78 and 280.01 for y_1 and 32.39 and 252.06 for y_2 respectively. Thus, the box objective of B_0 is calculated as:

$$Obj_{B_0} = \{(35.78 - T_1)^2 + 280.01\} + \{(32.39 - T_2)^2 + 252.06\} = 1428.87$$

Subsequent steps were conducted according to the procedure explained in Section 3. A total of 12 iterations of Steps 2, 3, and 4 were conducted as shown in Table 3.

Box B_{12} has the smallest box objective value and so was selected as the optimal box. The mean of responses for B_{12} was 53.13 for y_1 and 48.08 for y_2 which are quite close to the target values of 60 and 50, respectively. The box objective of B_{12} is 222.08, which is significantly smaller than that of B_0 .

In order to avoid over-fitting problems, we conducted the cross validation as in Section 4.2. Table 4 reports the box objective values from the data of training and validation sets. The box objective from the validation set is larger than the training set; however, the difference is not significant. This indicates that the obtained optimal box is not over-fitted.

5.2. Trade-Off Between Mean and Variance in EMR-PRIM

The mean and variance of the response variables are often in conflict. In this section, we run the EMR-PRIM with different weights in order to investigate the relationship between the mean and variance. We use the weighted MSE (WMSE) as a new box objective. According to Ding et al. (2004), using WMSE as an optimization criterion is a good way to balance the mean and variance. The new box objective is as follows:

$$Obj_B = \lambda (\bar{Y}_{1,B} - T_1)^2 + (1 - \lambda) s_{1,B}^2 + \lambda (\bar{Y}_{2,B} - T_2)^2 + (1 - \lambda) s_{2,B}^2$$

where $\bar{y}_{1,B}$ and $\bar{y}_{2,B}$ are the means of y_1 and y_2 of the observations located in box B , respectively. $s_{1,B}^2$ and $s_{2,B}^2$ are the sample variances of y_1 and y_2 of the observations located in box B , respectively. λ is a weighting factor ranging from 0 to 1. We set λ to 0.33 and 0.67. Table 5 shows the result of the optimization according to λ value.

The result of the optimization of Case 1 is the same as that in Table 4. In Case 2, the means of the two responses became worse compared to those of Case 1 due to the smaller λ value (i.e.,

Table 3. Optimization results for the case example obtained by EMR-PRIM

Iteration	Training Set					
	β_B	Box Obj.	$\bar{y}_1$	s_1^2	$\bar{y}_2$	s_2^2
1	0.63	1172.72	39.73	294.49	35.65	261.59
2	0.40	941.52	43.59	289.03	39.02	262.57
3	0.35	764.27	46.32	262.78	41.51	242.29
4	0.31	657.95	48.39	246.90	43.36	232.19
5	0.28	553.68	50.50	224.83	45.23	215.98
6	0.25	485.25	52.07	206.07	46.67	205.14
7	0.23	434.12	53.58	195.79	48.07	193.34
8	0.21	389.90	55.06	182.71	49.52	182.60
9	0.18	351.59	56.51	166.80	51.17	171.25
10	0.07	310.19	51.22	112.39	45.56	101.08
11	0.06	266.16	52.55	108.17	46.95	93.20
12	0.05	222.08	53.13	89.95	48.08	81.26

Table 4. Results of training and validation sets for the optimal box

Training Set					Validation Set				
Box obj.	y_1	s_1^2	$\bar{y}_2$	s_2^2	Box obj.	y_1	s_1^2	$\bar{y}_2$	s_2^2
222.08	53.13	89.95	48.08	81.26	457.65	47.91	122.43	42.78	137.03

Table 5. Mean and variance with different values of the weight factor λ

Case	λ	$\bar{y}_1$	s_1^2	$\bar{y}_2$	s_2^2
1	0.50	53.13	89.95	48.08	81.26
2	0.33	52.62	80.44	46.82	85.43
3	0.67	60.53	119.01	54.34	134.59

52.62 < 53.12 and 46.82 < 48.08). In contrast, the sum of the two variances decreased compared to Case 1 (i.e., 80.44 + 85.43 < 89.95 + 81.26) because a low λ value gives more weight to variance terms. On the contrary to Case 2, Case 3 gives more weight to the mean terms. Thus, the WMSE-based approach can help process engineers investigate tradeoffs between the mean and variance of a response.

6. EXAMPLE 3: CONCRETE COMPRESSIVE STRENGTH PROBLEM WITH THREE RESPONSES

In this section, we conduct a case example with three responses in order to support the validity of the proposed method for multiple responses. We arbitrarily generate y_3 values that have a large scale than y_1 and y_2 values. The target values of the three responses are 60 ($= T_1$), 50 ($= T_2$) and 170 ($= T_3$), respectively. The purpose of this case study is to maintain the means of y_1 , y_2 and y_3 at their target values, while minimizing the standard deviation of responses by controlling the input variables.

In Step 1, initial parameters are same as described in Section 5. The mean and variance of observation responses are 35.78 and 280.01 for y_1 , 32.39 and 252.06 for y_2 , 143.32 and 5340.33 for y_3 , respectively. Thus, the box objective of B_0 is calculated as:

$$Obj_{B_0} = \{(35.78 - T_1)^2 + 280.01\} + \{(32.39 - T_2)^2 + 252.06\} + \{(143.32 - T_3)^2 + 5340.33\} = 7481.27$$

As a result, a total of 16 iterations of Steps 2, 3, and 4 were conducted.

The second row of Table 6 shows the optimization result. The mean and variance of the optimal box for y_3 is much better than the box B_0 whereas y_1 and y_2 are not. In other words, it is difficult to improve all of responses from this box objective (i.e., Equation (1)) because the optimization is mainly driven by a response whose scale is relatively large. Hence, we used the standardized box objective for considering the difference scales of the three responses in the optimization procedure:

$$Obj_B = \frac{(\bar{Y}_{1,B} - T_1)^2 + s_{1,B}^2}{280.01} + \frac{(\bar{Y}_{2,B} - T_2)^2 + s_{2,B}^2}{252.06} + \frac{(\bar{Y}_{3,B} - T_3)^2 + s_{3,B}^2}{5340.33}$$

The third row of Table 6 shows the optimization result with standardized box objective. Although the variance of y_3 become worse, the mean and variance of y_1 and y_2 become significantly better than the previous box objective. In this regard, we suggest to use the standardized box objective when the scales of the responses are different.

Table 6. Comparison of the optimal box in Example 2 and 3

Example	$\mathbf{x}^*$	y_1		y_2		y_3	
		$\bar{y}_1$	s_1^2	$\bar{y}_2$	s_2^2	$\bar{y}_3$	s_3^2
Example 2	([213.7, 491], [24, 262.2], [0, 195], [121.75, 246.9], [3.9, 18.6], [801, 1134.3], [594, 992.6], [56, 100])	53.13	89.95	48.08	81.26		
Example 3	([144, 387], [19, 190], [87, 148], [154, 192], [7, 19], [814, 953.3], [665.6, 845], [3, 56])	42.67	216.92	37.44	152.87	162.32	2689.08
Example 3 with standardized box obj.	([264.5, 446], [17.2, 180], [24.46, 163], [145, 209.7], [5.9, 22.1], [822, 1061.7], [633, 871], [28, 100])	52.73	112.47	46.40	105.29	158.24	3914.65

7. CONCLUSION

In this work, we proposed a modified version of PRIM, called EMR-PRIM, in an attempt to optimize the mean and standard deviation of multiple responses. The method obtains an optimal setting of input variables directly from the operational data. Thus, no experiments are required, and it is free from the uncertainty issues resulting from model building. We illustrated EMR-PRIM with a case example in Section 5 and additionally analyzed the tradeoff between mean and variance. It was shown that EMR-PRIM works well and has advantages in finding the optimal box to solve the MRO problem.

We suggest three future research topics. First, the effect of box objective function requires investigation. In this paper, we employed a particular box objective, MSE. However, other indices, such as desirability function (Derringer & Suich, 1980; Kim & Lin, 1998; Kim & Lin, 2006; Jeong & Kim, 2009) and loss function (Pignatiello, 1993; Vining, 1998; Ko et al., 2005), are available. The desirability function has been widely used for optimizing means of multiple responses; however, it can also be used for the simultaneous optimization of mean and variance of multiple responses. The loss function also can be used for this purpose. The loss function has an advantage in that it conveys covariance of multiple responses. Secondly, WMSE can be revised to give weights for each response. In the current WMSE, the weight factor assigns relative weights to the mean and variance. In some cases, some responses are more important than others. In such a case, different weights should be assigned to the multiple responses. Thirdly, clustering algorithms can be considered in reducing the search space. By conducting clustering, the entire observations are divided into several clusters. Then, the proposed PRIM approach can be applied to each cluster. This approach can be considered as “divide and conquer” which is effective in improving the efficiency of the proposed method. Taniar et al. (2008) and Kwot et al. (2002)’s approach which deal with the mining algorithm and clustering algorithm can be applied in this approach.

ACKNOWLEDGMENT

This work was supported by a National Research Foundation of Korea (NRF) grant funded by the Korean government (MSIP) (No. 2015R1C1A1A01051952).

REFERENCES

- Chong, I., Albin, S., & Jun, C. (2007). A data mining approach to process optimization without an explicit quality function. *IIE Transactions*, 39(8), 795–804. doi:10.1080/07408170601142668
- Derringer, G., & Suich, R. (1980). Simultaneous optimization of several response variables. *Journal of Quality Technology*, 12(4), 214–219.
- Ding, R., Lin, D., & Wei, D. (2004). Dual-Response Surface Optimization: A Weighted MSE Approach. *Quality Engineering*, 16(3), 377–385. doi:10.1081/QEN-120027940
- Friedman, J., & Fisher, N. (1999). Bump hunting in high-dimensional data. *Statistics and Computing*, 9(2), 123–143.
- Jeong, I., & Kim, K. (2009). An interactive desirability function method to multiresponse optimization. *European Journal of Operational Research*, 195(2), 412–426. doi:10.1016/j.ejor.2008.02.018
- Jeong, I., Kim, K., & Chang, S. (2005). Optimal Weighting of Bias and Variance in Dual Response Surface Optimization. *Journal of Quality Technology*, 37(3), 236–247.
- Kim, K., & Lin, D. (1998). Dual response surface optimization: A fuzzy modeling approach. *Journal of Quality Technology*, 30(1), 1–10.
- Kim, K., & Lin, D. (2006). Optimization of multiple responses considering both location and dispersion effects. *European Journal of Operational Research*, 169(1), 133–145. doi:10.1016/j.ejor.2004.06.020
- Ko, Y., Kim, K., & Jun, C. (2005). A new loss function-based method for multiresponse optimization. *Journal of Quality Technology*, 37(1), 50–59.
- Kwok, T., Smith, K., Lozano, S., & Taniar, D. (2002). Parallel fuzzy c-means clustering for large data sets. In *Proceedings of the Euro-Par 2002 parallel processing* (pp. 27–58).
- Lee, D., Jeong, I., & Kim, K. (2010). A Posterior Preference Articulation Approach to Dual-Response Surface Optimization. *IIE Transactions*, 42(2), 161–171. doi:10.1080/07408170903228959
- Lee, D., & Kim, K. (2012). Interactive Weighting of Bias and Variance in Dual Response Surface Optimization. *Expert Systems with Applications*, 39(5), 5900–5906. doi:10.1016/j.eswa.2011.11.114
- Lee, D., & Kim, K. (2013). Determining the target value of ACICD to optimize the electrical characteristics of semiconductors using dual response surface optimization. *Applied Stochastic Models in Business and Industry*, 29(4), 377–386. doi:10.1002/asmb.1973
- Lee, D., Yang, J., & Kim, K. (2017). Dual-Response Optimization Using a Patient Rule Induction Method. Working paper.
- Lee, H., & Lee, D. (2016). A solution selection approach to multiresponse surface optimization based on a clustering method. *Quality Engineering*, 28(4), 388–401. doi:10.1080/08982112.2016.1155222
- Lee, M., & Kim, K. (2008). MR-PRIM: Patient Rule Induction Method for Multiresponse Optimization. *Quality Engineering*, 20(2), 232–242. doi:10.1080/08982110701778468
- Lin, D., & Tu, W. (1995). Dual response surface optimization. *Journal of Quality Technology*, 27(1), 34–39.
- Myers, R., & Carter, W. (1973). Response Surface Techniques for Dual Response Systems. *Technometrics*, 15(2), 301–317. doi:10.1080/00401706.1973.10489044
- Myers, R., Montgomery, D., & Anderson-Cook, C. (2009). *Response Surface Methodology*. New Jersey: Wiley & Sons.
- Palmer, K., Asiabanpour, B., & Khoshnevis, B. (2006). Development of a rapid prototyping system using response surface methodology. *Quality and Reliability Engineering International*, 22(8), 919–937. doi:10.1002/qre.779
- Pignatiello, J. Jr. (1993). Strategies for robust multiresponse quality engineering. *IIE Transactions*, 25(3), 5–15. doi:10.1080/07408179308964286

- Reddy, P., Nishina, K., & Babu, S. (1997). Unification of robust design and goal programming for multiresponse optimization: A case study. *Quality and Reliability Engineering International*, 13(6), 371–383. doi:10.1002/(SICI)1099-1638(199711/12)13:6<371::AID-QRE127>3.0.CO;2-Z
- Taniar, D., Rahayu, W., Lee, V., & Daly, O. (2008). Exception rules in association rule mining. *Applied Mathematics and Computation*, 205(2), 735–750. doi:10.1016/j.amc.2008.05.020
- Vining, G. (1998). A compromise approach to multiresponse optimization. *Journal of Quality Technology*, 30(4), 309–313.
- Vining, G., & Myers, R. (1990). Combining Taguchi and response surface philosophies: A dual response approach. *Journal of Quality Technology*, 22(1), 38–45.
- Xu, D., & Albin, S. (2003). Robust Optimization of Experimentally Derived Objective Functions. *IIE Transactions*, 35(9), 793–802. doi:10.1080/07408170304408
- Yeh, I. (1998). Modeling of strength of high-performance concrete using artificial neural networks. *Cement and Concrete Research*, 28(12), 1797–1808. doi:10.1016/S0008-8846(98)00165-3

Jin-Kyung Yang is a M.S candidate student in the Division of Interdisciplinary Industrial Studies at Hanyang University in Korea. Her research includes statistical quality control, design of experiments, and response surface methodology and so on.

Dong-Hee Lee is an assistant professor in the Division of Interdisciplinary Industrial Studies at Hanyang University in Korea. He received his Ph.D. in Industrial and Management Engineering in 2011 from Pohang University of Science and Technology (Korea). He worked as a senior researcher in quality team of semiconductor division at Samsung Electronics for four years and received CRE (certified reliability engineer) from ASQ (American Society for Quality). His research interests include quality improvement methods such as response surface methodology, design of experiments, statistical process control, Six Sigma and so on.